

EXP-1

The RStudio interface for EXP-1 shows a script with the following code:

```
1 a <- as.numeric(readline(prompt = "Enter first number: "))
2 b <- as.numeric(readline(prompt = "Enter second number: "))
3
4 sum <- a + b
5
6 print(paste("Sum =", sum))
```

The console output shows the execution of the script, including prompts for input and the resulting sum:

```
[1] "Greatest number is 10"
> a <- as.numeric(readline(prompt = "Enter first number: "))
Enter first number: 10
> b <- as.numeric(readline(prompt = "Enter second number: "))
Enter second number: 20
> sum <- a + b
>
> print(paste("Sum =", sum))
[1] "Sum = 30"
```

The Environment pane shows the following values:

Variable	Value
a	10
b	20
c	6
sum	30

The Files pane shows a list of files in the Home directory, including .Rhistory, 192372057_1_docx, Arduino, burp-history.xml, burpfile, Chromium, CO2_AT-1.docx, CSA15_CO1_AT3_Technical_Presentati..., Custom Office Templates, desktop.ini, EXP1.C, EXP1.vavgul_HP.bak, EXP1.vavgul.HP, EXP1.vavgul, EXP1.vavgulproj, Lab-1.pdpsprj, and Lab-1.pdpsprj.

EXP-2

The RStudio interface for EXP-2 shows a script with the following code:

```
1 a <- as.numeric(readline(prompt = "Enter first number: "))
2 b <- as.numeric(readline(prompt = "Enter second number: "))
3
4 sub <- a - b
5
6 print(paste("Sub =", sub))
```

The console output shows the execution of the script, including prompts for input and the resulting sum:

```
[1] "Greatest number is 10"
> a <- as.numeric(readline(prompt = "Enter first number: "))
Enter first number: 10
> b <- as.numeric(readline(prompt = "Enter second number: "))
Enter second number: 20
> sum <- a + b
>
> print(paste("Sum =", sum))
[1] "Sum = 30"
> a <- as.numeric(readline(prompt = "Enter first number: "))
Enter first number: 10
> b <- as.numeric(readline(prompt = "Enter second number: "))
Enter second number: 5
> sub <- a - b
>
> print(paste("Sub =", sub))
[1] "Sub = 5"
```

The Environment pane shows the following values:

Variable	Value
a	10
b	5
c	6
sub	5
sum	30

The Files pane shows a list of files in the Home directory, including .Rhistory, 192372057_1_docx, Arduino, burp-history.xml, burpfile, Chromium, CO2_AT-1.docx, CSA15_CO1_AT3_Technical_Presentati..., Custom Office Templates, desktop.ini, EXP1.C, EXP1.vavgul_HP.bak, EXP1.vavgul.HP, EXP1.vavgul, EXP1.vavgulproj, Lab-1.pdpsprj, and Lab-1.pdpsprj.

EXP-3

RStudio interface showing the execution of an R script for EXP-3. The script prompts the user to enter two numbers and calculates their multiplication.

```
1 a <- as.numeric(readline(prompt = "Enter first number: "))
2 b <- as.numeric(readline(prompt = "Enter second number: "))
3
4 mul <- a * b
5
6 print(paste("Multiplication =", mul))
```

Console output:

```
R - R4.6.1 ~ /
> # R script: exp3.R
> # Prompt user for two numbers and calculate multiplication
> Enter first number: 10
> Enter second number: 20
> sum <- a + b
>
> print(paste("Sum =", sum))
[1] "Sum = 30"
> a <- as.numeric(readline(prompt = "Enter first number: "))
> Enter first number: 10
> b <- as.numeric(readline(prompt = "Enter second number: "))
> Enter second number: 5
> sub <- a - b
>
> print(paste("Sub =", sub))
[1] "Sub = 5"
> mul <- a * b
>
> print(paste("Multiplication =", mul))
[1] "Multiplication = 50"
> |
```

Environment pane shows the following values:

Variable	Value
a	10
b	5
c	6
mul	50
sub	5
sum	30

EXP-4

RStudio interface showing the execution of an R script for EXP-4. The script prompts the user to enter two numbers and calculates their division.

```
1 a <- as.numeric(readline(prompt = "Enter first number: "))
2 b <- as.numeric(readline(prompt = "Enter second number: "))
3
4 Div <- a / b
5
6 print(paste("Division =", Div))
```

Console output:

```
R - R4.6.1 ~ /
> # R script: exp4.R
> # Prompt user for two numbers and calculate division
> Enter first number: 10
> Enter second number: 5
> sub <- a - b
>
> print(paste("Sub =", sub))
[1] "Sub = 5"
> mul <- a * b
>
> print(paste("Multiplication =", mul))
[1] "Multiplication = 50"
> Div <- a / b
>
> print(paste("Division =", Div))
[1] "Division = 2"
> |
```

Environment pane shows the following values:

Variable	Value
a	10
b	5
c	6
Div	2
mul	50
sub	5
sum	30

EXP-5

The screenshot shows the RStudio interface with a script editor containing the following code:

```
1 num=as.integer(readline(prompt="enter a number:"))
2 if((num%2)==0)
3 {
4   print("number is a even")
5 } else{
6   print("number is odd")
7 }
8
```

The console output shows the execution of the script with the input 50:

```
R - R4.6.1 ~ /
> mul <- a * b
>
> print(paste("Multiplication =", mul))
[1] "Multiplication = 50"
> Div <- a / b
>
> print(paste("Division =", Div))
[1] "Division = 2"
> num=as.integer(readline(prompt="enter a number:"))
enter a number:5
> if((num%2)==0)
+ {
+   print("number is a even")
+ } else{
+   print("number is odd")
+ }
[1] "number is odd"
>
```

The Environment pane shows the following variables:

Variable	Value
a	10
b	5
c	6
Div	2
mul	50
num	5L
sub	5
sum	30

The Files pane shows the following files:

Name	Size	Modified
.Rhistory	0 B	Aug 8, 2026, 9:50 AM
192372057_1_docx	92.8 KB	Jun 7, 2024, 9:25 AM
Arduino		
burp-history.xml	8.3 MB	Jun 23, 2026, 2:18 PM
burpfile	8.3 MB	Jun 23, 2026, 2:21 PM
Chromium		
CO2_AT-1.docx	2814 KB	Jul 29, 2026, 12:57 PM
CSA15_CO1_AT3_Technical_Presentati...	303 KB	Jul 20, 2026, 2:46 PM
Custom Office Templates		
desktop.ini	418 B	Aug 3, 2026, 7:37 PM
EXP1.C	0 B	Aug 4, 2026, 2:45 PM
EXP1.vngul_HP.bak	68.9 KB	Aug 4, 2026, 2:45 PM
EXP1.vngul_HP	68.9 KB	Aug 4, 2026, 2:45 PM
EXP1.vuopt	892 B	Aug 4, 2026, 2:45 PM
EXP1.vuproj	580 B	Aug 4, 2026, 2:45 PM
Lab-1.pdpsrj	15.6 KB	Jul 9, 2026, 11:56 AM
Lab-1.vngul_HP	6.6 KB	Jul 9, 2026, 11:56 AM

EXP-6

The screenshot shows the RStudio interface with a script editor containing the following code:

```
1 names<-c("siri","mahi","chiru")
2 age<-c(23,24,25)
3 marks<-c(88,78,25)
4 df<-data.frame(names,age,marks)
5 mean(df $age)
6 write.csv(df,"datafr.csv")
7 median(df $age)
8 write.csv(df,"datafr.csv")
9 median(df $age)
10 write.csv(df,"datafr.csv")
11
```

The console output shows the execution of the script:

```
R - R4.6.1 ~ /
> print("number is odd")
[1] "number is odd"
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> mean(df $age)
[1] 24
> write.csv(df,"datafr.csv")
[1] 24
> median(df $age)
[1] 24
> write.csv(df,"datafr.csv")
[1] 24
> median(df $age)
[1] 24
> write.csv(df,"datafr.csv")
>
```

The Environment pane shows the following variables:

Variable	Value
df	3 obs. of 3 variables
a	10
age	num [1:3] 23 24 25
b	5
c	6
Div	2
marks	num [1:3] 88 78 25
mul	50
names	chr [1:3] "siri" "mahi" "chiru"

The Files pane shows the following files:

Name	Size	Modified
.Rhistory	0 B	Aug 8, 2026, 9:50 AM
192372057_1_docx	92.8 KB	Jun 7, 2024, 9:25 AM
Arduino		
burp-history.xml	8.3 MB	Jun 23, 2026, 2:18 PM
burpfile	8.3 MB	Jun 23, 2026, 2:21 PM
Chromium		
CO2_AT-1.docx	2814 KB	Jul 29, 2026, 12:57 PM
CSA15_CO1_AT3_Technical_Presentati...	303 KB	Jul 20, 2026, 2:46 PM
Custom Office Templates		
datafr.csv	81 B	Aug 8, 2026, 10:40 AM
desktop.ini	418 B	Aug 3, 2026, 7:37 PM
EXP1.C	0 B	Aug 4, 2026, 2:45 PM
EXP1.vngul_HP.bak	68.9 KB	Aug 4, 2026, 2:45 PM
EXP1.vngul_HP	68.9 KB	Aug 4, 2026, 2:45 PM
EXP1.vuopt	892 B	Aug 4, 2026, 2:45 PM
EXP1.vuproj	580 B	Aug 4, 2026, 2:45 PM
Lab-1.pdpsrj	15.6 KB	Jul 9, 2026, 11:56 AM
Lab-1.vngul_HP	6.6 KB	Jul 9, 2026, 11:56 AM

EXP-7

The RStudio interface displays a script in the editor pane with the following code:

```
1 names<-c("siri","mahi","chiru")
2 age<-c(23,24,25)
3 marks<-c(88,78,25)
4 df<-data.frame(names,age,marks)
5 summary(df $age)
6 write.csv(df,"datafr.csv")
7
```

The console shows the execution of the script:

```
R - R4.6.1 - ~/R
> mean(df $age)
[1] 24
> write.csv(df,"datafr.csv")
> median(df $age)
[1] 24
> write.csv(df,"datafr.csv")
> median(df $age)
[1] 24
> write.csv(df,"datafr.csv")
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> summary(df $age)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 23.0   23.5   24.0   24.0   24.5   25.0
> write.csv(df,"datafr.csv")
>
```

The Environment pane shows the data frame 'df' with 3 observations and 3 variables:

Variable	Value
a	10
age	num [1:3] 23 24 25
b	5
c	6
Div	2
marks	num [1:3] 88 78 25
mul	50
names	chr [1:3] "siri" "mahi" "chiru"

The Files pane shows the file 'datafr.csv' in the current directory.

EXP-8

The RStudio interface displays a script in the editor pane with the following code:

```
1 a <- as.numeric(readline(prompt = "Enter first number: "))
2 b <- as.numeric(readline(prompt = "Enter second number: "))
3 c <- as.numeric(readline(prompt = "Enter third number: "))
4
5 if (a >= b && a >= c) {
6   print(paste("Greatest number is", a))
7 } else if (b >= a && b >= c) {
8   print(paste("Greatest number is", b))
9 } else {
10  print(paste("Greatest number is", c))
11 }
```

The console shows the execution of the script:

```
R - R4.6.1 - ~/R
WARNING message:
NAS introduced by coercion
> a <- as.numeric(readline(prompt = "Enter first number: "))
Enter first number: 10
> b <- as.numeric(readline(prompt = "Enter second number: "))
Enter second number: 5
> c <- as.numeric(readline(prompt = "Enter third number: "))
Enter third number: 6
> if (a >= b && a >= c) {
+   print(paste("Greatest number is", a))
+ } else if (b >= a && b >= c) {
+   print(paste("Greatest number is", b))
+ } else {
+   print(paste("Greatest number is", c))
+ }
[1] "Greatest number is 10"
>
```

The Environment pane shows the variables 'a', 'b', and 'c' with values 10, 5, and 6 respectively.

EXP-9

RStudio interface showing the execution of R code to create a data frame and calculate summary statistics.

```
1 names<-c("siri","mahi","chiru")
2 age<-c(23,24,25)
3 marks<-c(88,78,25)
4 df<-data.frame(names,age,marks)
5 IQR(df $age)
6 write.csv(df,"datafr.csv")
7
```

Console output:

```
R - R4.6.1 ~ /...
> write.csv(df,"datafr.csv")
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> summary(df $age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    23.0   23.5   24.0   24.0   24.5   25.0
> write.csv(df,"datafr.csv")
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> IQR(df $age)
[1] 1
> write.csv(df,"datafr.csv")
>
```

Environment pane shows the data frame 'df' with 3 observations and 3 variables:

a	10	
age	num [1:3]	23 24 25
b	5	
c	6	
Div	2	
marks	num [1:3]	88 78 25
mul	50	
names	chr [1:3]	"siri" "mahi" "chiru"

EXP-10

RStudio interface showing the execution of R code to create a data frame and calculate quantiles.

```
1 names<-c("siri","mahi","chiru")
2 age<-c(23,24,25)
3 marks<-c(88,78,25)
4 df<-data.frame(names,age,marks)
5 quantile(df $age)
6 write.csv(df,"datafr.csv")
7
```

Console output:

```
R - R4.6.1 ~ /...
> write.csv(df,"datafr.csv")
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> quantile(df $age)
 0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df,"datafr.csv")
>
```

Environment pane shows the data frame 'df' with 3 observations and 3 variables:

a	10	
age	num [1:3]	23 24 25
b	5	
c	6	
Div	2	
marks	num [1:3]	88 78 25
mul	50	
names	chr [1:3]	"siri" "mahi" "chiru"

EXP-11

The screenshot shows the RStudio interface. The script editor contains the following code:

```
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 mid <- mean(range(df$age))
8
9 write.csv(df, "datafr.csv")
```

The console shows an error message: "Error: unexpected symbol in \"mid range\"". The environment pane on the right shows the following variables:

Variable	Value
a	10
age	num [1:3] 23 24 25
b	5
c	6
Div	2
marks	num [1:3] 88 78 25
mid	24
mul	50
names	chr [1:3] "siri" "mahi" "chiru"

EXP-12

The screenshot shows the RStudio interface. The script editor contains the following code:

```
1 library(readxl)
2
3 diabetest1 <- read_excel("C:/Users/HP/Downloads/NARA_example.xlsx")
4 A <- diabetest1$Age
5 Mean <- mean(A)
6 Std <- sd(A)
7 Zscore <- (A - Mean) / Std
8
9 Zscore
```

The console shows an error message: "Error: 'U' used without hex digits in character string (<input>:1:30)". The environment pane on the right shows the following variables:

Variable	Value
diabetest1	10 obs. of 1 variable
a	10
A	num [1:10] 23 24 25 22 28 30 26 27 21 29
age	num [1:3] 23 24 25
b	5
c	6
Div	2
marks	num [1:3] 88 78 25
Mean	25.5

EXP-13

RStudio interface showing a script with the following code:

```
1 diabetest1 <- read_excel("C:/Users/HP/Downloads/question_13_MIN_MAX_MEAN_MINMAX.xlsx")
2
3 A <- as.numeric(diabetest1$Age)
4 Mean <- mean(A)
5
6 Mean
```

The console shows a warning message:

```
Warning message:
In mean.default(A) : argument is not numeric or logical: returning NA
```

The environment pane shows the following data frame:

diabetest1	63 obs. of 2 variables
a	10
A	num [1:63] 21 22 23 24 25 26 27 28 29 30...
age	num [1:3] 23 24 25
b	5
c	6
Div	2
marks	num [1:3] 88 78 25
Mean	NA_real_

EXP-14

RStudio interface showing a script with the following code:

```
1 a<-c(55,67,89,80,90)
2 barplot(a)
3 a<-c(55,67,89,80,90)
4 barplot(a)
5 barplot(a,horiz=TRUE)
6
```

The console shows a warning message:

```
Warning message:
NAs introduced by coercion
```

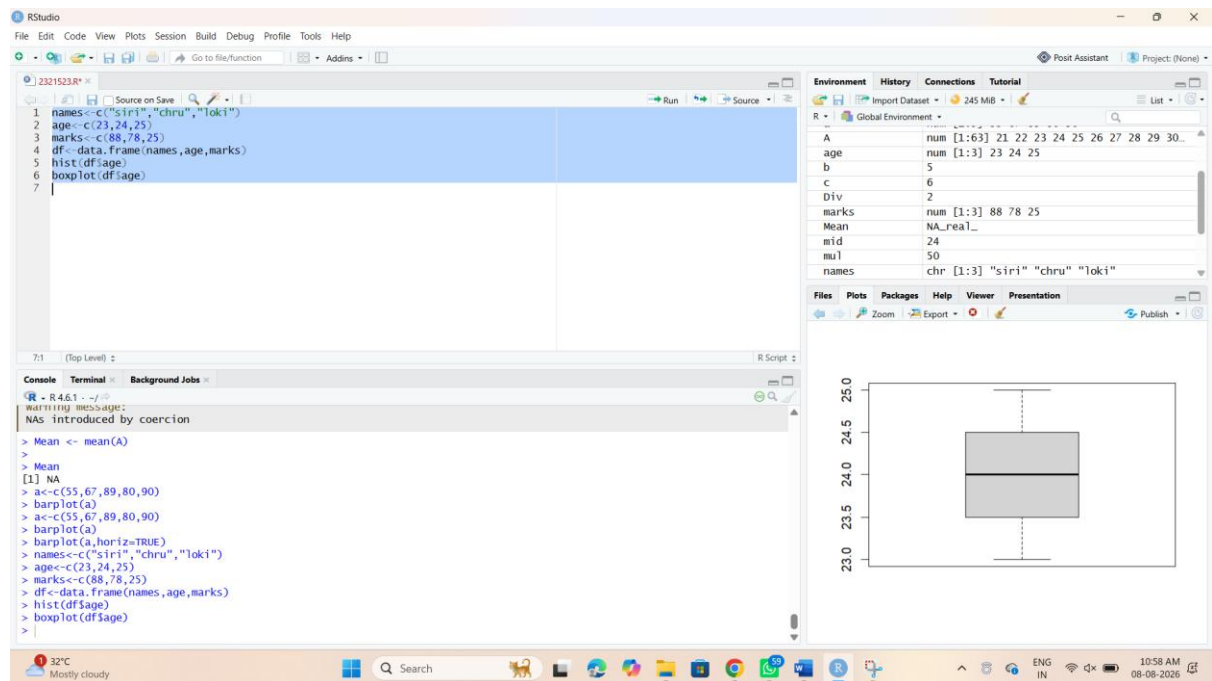
The environment pane shows the following data frame:

diabetest1	63 obs. of 2 variables
a	num [1:5] 55 67 89 80 90
A	num [1:63] 21 22 23 24 25 26 27 28 29 30...
age	num [1:3] 23 24 25
b	5
c	6
Div	2
marks	num [1:3] 88 78 25
Mean	NA_real_

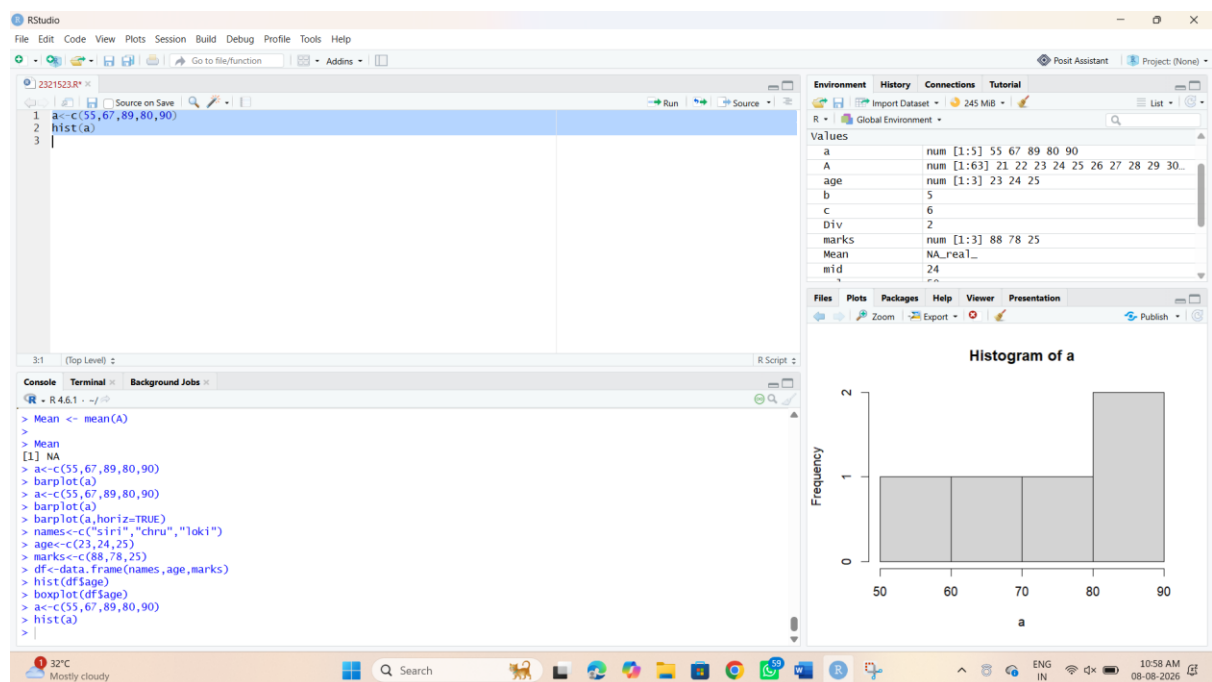
The plot shows a horizontal bar chart with the following data:

Variable	Value
a	55
A	67
age	89
b	80
c	90

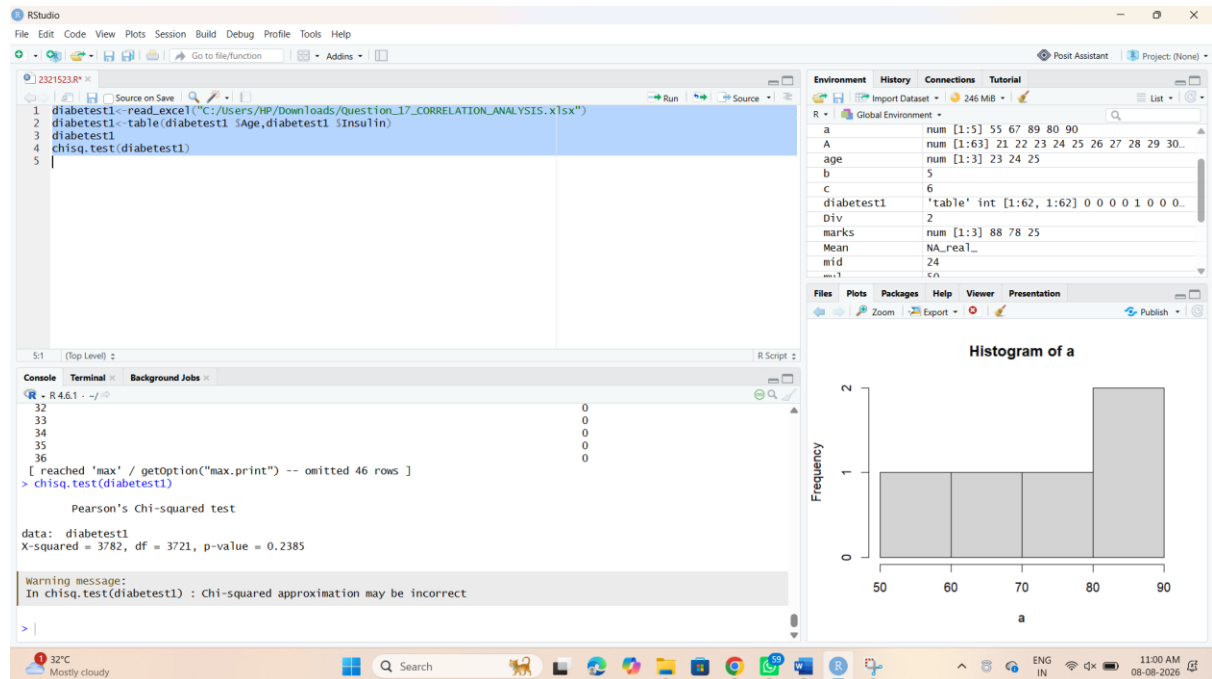
EXP-15



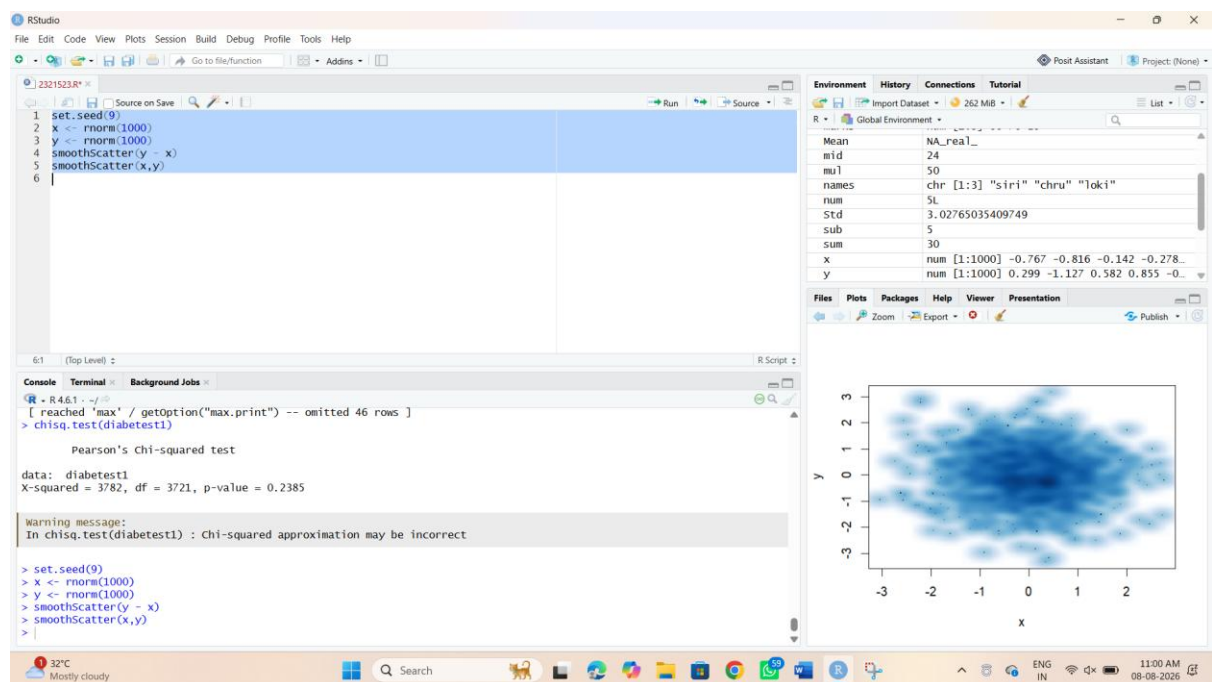
EXP-16



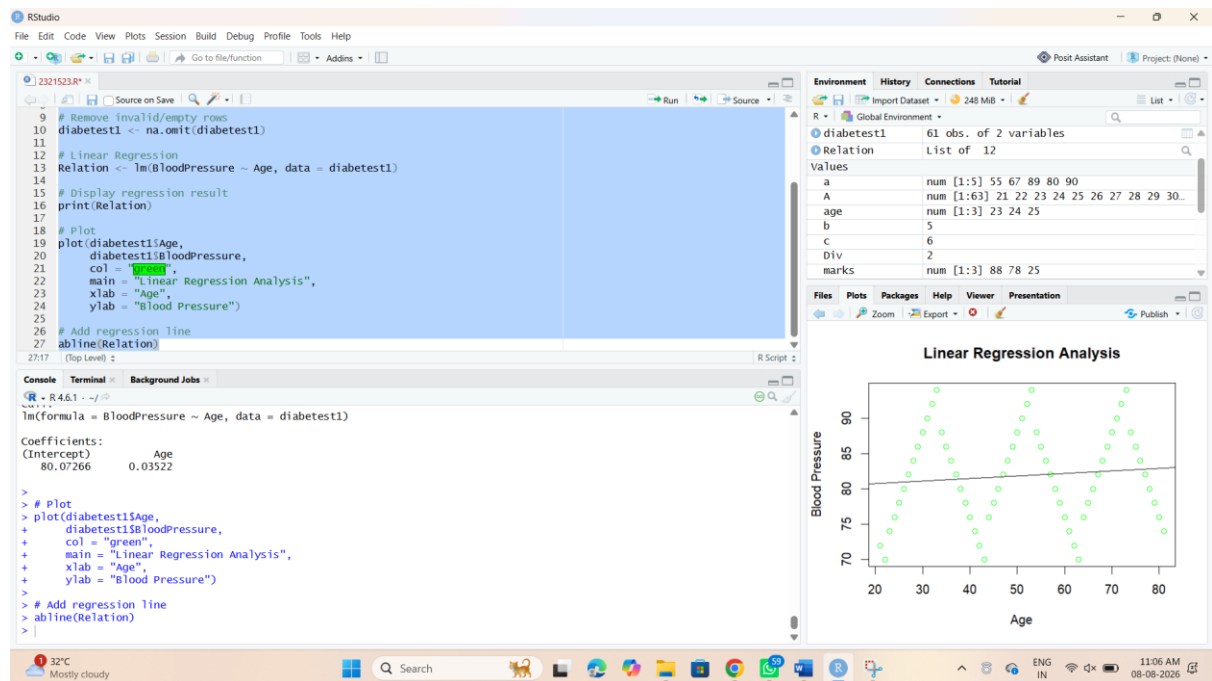
EXP-17



EXP-18



EXP-19



EXP-20

